

POFM — Prediction Output Formation Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Formation Standard (PFOS)

Operational Layer: Prediction Formation Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Prediction Output Formation

Version: 1.0

Status: Canonical · Open Module

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Prediction Output Formation

Canonical Definition

Prediction Output Formation Module (POFM) defines the governance conditions under which an Identified Predictive Claim is transformed into a specific, bounded, attributable, machine- and human-readable Prediction Output that preserves the claim's meaning, conditions, horizon, scope, and formation context.

POFM governs the formation of the explicit output representation through which the Predictive Claim becomes operationally usable within the Prediction Formation process.



transforms:

Governed Prediction Composition + Identified Predictive Claim + Applicable Formation Logic

into:

Governed Prediction Output

POFM does not determine the overall formation state of the prediction.

That belongs to:

PSEM — Prediction State Establishment Module.

Its foundational distinction is:

Predictive Claim $\neq$ Prediction Output Representation

Operational Role

POFM serves as the third internal governance space of the Prediction Formation Standard (PFOS) .

It receives:

Governed Prediction Composition

Identified Predictive Claim

Applicable Formation Logic

and establishes the Prediction Output that expresses that claim in the form required for further prediction governance.

The module asks:

What exact output represents this Predictive Claim, and does that output preserve the meaning, scope, horizon, conditions, and provenance of the claim from which it was formed?

The governed progression is:

Identified Predictive Claim → Output Form Selection → Output Structure Definition → Value / Category / Range Formation → Condition Preservation → Semantic Integrity Check → Output State → Governed Prediction Output

The primary output is:

Governed Prediction Output

This becomes a principal input to:

PSEM — Prediction State Establishment Module.

Internal Module Flow

Prediction Composition → Predictive Claim Identification → Prediction Output Formation → Prediction State Establishment → Prediction Formation Traceability

Current Position: Prediction Output Formation

Module Operational Space

POFM governs:

Prediction Output identification; output form; output structure; output value; output category; output range; output threshold; output direction; output timing; output unit; output baseline; output conditionality; output scenario association; output semantics; output normalization; output transformation; output selection; output aggregation; output adjustment; output precision; output rounding; output completeness; output consistency; output-human readability; output-machine readability; output interoperability; output provenance; output source;

output derivation; output duplication; output conflict; output state; output version; and historical output integrity.

Module Function

A Predictive Claim may be clear while its final output representation remains poorly governed.

Example:

Predictive Claim:

Demand for Product A in Region X is predicted to exceed 10,000 units during Q4.

Possible outputs may include:

10,800

> 10,000

High Demand

82%

Q4 Demand Increase Expected

These outputs are not equivalent.

Some represent:

a point estimate; a threshold; a category; a probability; a qualitative summary.

POFM determines which output legitimately represents the Predictive Claim and preserves the relationship between the two.

Prediction Output

A Prediction Output is the explicit representation produced by the Prediction Formation process that expresses a specific Predictive Claim in a form suitable for further governance, interpretation, system use, or later comparison.

A Prediction Output may be:

numeric; categorical; directional; interval-based; threshold-based; temporal; event-based; qualitative; structured; machine-readable; composite.

Prediction Output ≠ Raw Model Output

A raw model output may contribute to Prediction Formation.

It does not automatically become the Governed Prediction Output.

Example:

Raw model score:

0.81

Governed Prediction Output:

Event X is predicted to occur within 30 days, with the applicable probability representation governed separately under PPUS.

The output used by the model internally may differ from the final prediction representation.

Raw Model Output ≠ Governed Prediction Output

Output Form

The Output Form describes how the Predictive Claim is represented.

Possible forms include:

Point Estimate

Range

Threshold

Category

Direction

Event State

Timing

Conditional Output

Structured Multi-Field Output

The form must fit the claim.

Point Prediction Output

A Point Prediction Output represents one future value.

Example:

Predicted demand: 12,400 units

A point output should not imply greater certainty than the methodology supports.

Range Prediction Output

A Range Prediction Output represents a bounded interval.

Example:

Predicted demand: 11,000–13,000 units

The interval belongs to the Prediction Output.

PPUS later governs how uncertainty or probability associated with that range is represented.

Threshold Prediction Output

A Threshold Output represents whether a defined future value is predicted to cross a threshold.

Example:

Predicted state: Demand > 10,000 units during Q4

The threshold and direction must remain explicit.

Categorical Prediction Output

A prediction may be expressed through a governed category.

Example:

Predicted system risk state: High

The category must preserve the semantic meaning established under PCIM.

Directional Prediction Output

A Directional Output may state:

Increase

Decrease

Stable

where magnitude is not part of the governed claim.

Event Prediction Output

An Event Output represents predicted occurrence or non-occurrence.

Example:

Predicted event: Supply interruption occurs within 60 days

Timing Prediction Output

A Timing Output expresses when an event or state is predicted to occur.

Example:

Predicted project completion: Q2 2027

Conditional Prediction Output

Where the claim is conditional, the output must preserve the condition.

Example:

If supplier capacity remains above 80%, predicted output exceeds 50,000 units during Q4.

The output must not become:

Predicted output exceeds 50,000 units.

Conditional Claim $\neq$ Unconditional Output

Output Structure

A Governed Prediction Output may require multiple fields rather than one value.

For example:

Prediction Object: Product A Demand Predicted State: >10,000 units Prediction Horizon: Q4 2026

Boundary: Switzerland Condition: Supplier continuity remains Holding Formation Time: 8 September 2026

This structured representation may provide stronger governance than a standalone number.

Output Value

Where a numeric value is part of the claim, POFM governs the value actually carried into the formed Prediction Output.

The output value must remain distinguishable from:

raw model values; intermediate values; candidate values; adjusted values; scenario values.

Intermediate Output $\neq$ Final Prediction Output

A predictive pipeline may produce many intermediate values.

Example:

Raw Model: 13,240

↓

Calibration Layer: 12,980

↓

Analyst Adjustment: 12,500

↓

Governed Prediction Output: 12,500

Historical governance must preserve the material transformation path.

Output Selection

Where multiple candidate outputs exist, POFM may govern which one becomes the output representing the Identified Predictive Claim.

Selection should remain consistent with:

Prediction Methodology; Prediction Composition; Predictive Claim; applicable conditions.

Output Transformation

Prediction Output Formation may transform a predictive value.

Examples:

unit conversion; normalization; category mapping; threshold mapping; rounding; human adjustment; calibration-dependent conversion; formatting for machine interoperability.

Material transformation must remain traceable.

Transformed Output $\neq$ Original Output

Output Adjustment

A human or system process may adjust an output before final formation.

Example:

Model output: 8.6% growth

Expert-adjusted output: 7.5%

The final output may legitimately be 7.5%.

But the adjustment must not cause the 7.5% value to be falsely represented as the original model prediction.

Output Aggregation

Several component values may be combined to form one Prediction Output.

Example:

regional demand estimates may be aggregated into:

national demand prediction.

Aggregation must preserve:

boundary relationships; units; composition logic; applicable assumptions.

Output Normalization

Normalization may standardize prediction outputs for use across systems.

Examples:

YES / NO

0–1

Low / Medium / High

canonical date format

standard measurement unit

Normalization must preserve the predictive meaning.

Output Normalization $\neq$ Meaning Change

Output Unit

Numeric outputs must retain applicable units.

Examples:

CHF

units

percentage

megawatts

days

probability value

An output without its unit may become semantically misleading.

Numeric Value Without Unit $\neq$ Complete Numeric Prediction Output

Output Baseline

Where the output expresses change, the baseline must remain attached or reconstructable.

Example:

+8%

requires a reference such as:

compared with FY2026 revenue.

Output Horizon

The Prediction Horizon must remain part of the output context.

Example:

Demand: 12,000 units

without horizon may be meaningless.

Same Output Value + Different Horizon $\neq$ Same Prediction Output

Output Boundary

The applicable Predictive Boundary must remain preserved.

Example:

12,000 units

could refer to:

one store; one country; global sales.

The numeric value alone does not establish its governed meaning.

Output Conditions

Conditions materially affecting the Predictive Claim must remain associated with the output.

This may include:

assumptions; scenario; system state; policy state; threshold condition; operating condition.

Output Semantic Integrity

The Prediction Output must preserve the semantic meaning established under PCIM.

A claim about:

probability of threshold exceedance

must not be output as:

probability of system failure

unless those concepts are legitimately equivalent.

Output Representation $\neq$ Permission to Change Claim Meaning

Output Precision

Output precision must remain supported by the Prediction Methodology and Predictive Claim.

Example:

Methodology supports:

approximately 10–12 million

but output states:

10,873,421.53

This may create false precision.

Rounding

Rounding may be legitimate.

However, rounding must not materially change:

threshold position; category; decision relevance; later performance interpretation.

Example:

49.6% $\rightarrow$ 50%

may matter if 50% is a governance threshold.

Rounding $\neq$ Semantically Neutral by Default

Output Granularity

Output granularity should fit the Prediction Purpose.

Examples:

Strategic planning may require:

annual range

while real-time operations may require:

hourly point prediction.

Greater granularity is not automatically better.

Output Completeness

A Prediction Output is sufficiently complete where it carries or remains linked to all materially necessary context for its governed interpretation.

This may require:

output value/state; unit; horizon; boundary; condition; claim reference; prediction reference.

Standalone Output Sufficiency

Some outputs may be understandable only when used within a structured system.

A standalone number may be insufficient outside that system.

POFM therefore distinguishes:

display sufficiency

from:

governance sufficiency.

Human-Readable Output

Prediction Outputs intended for humans should preserve enough context to avoid foreseeable semantic distortion.

Example:

Instead of:

87%

use:

87% modeled probability of Event X within 30 days

where that is the governed meaning.

Machine-Readable Output

Prediction Outputs may require structured fields for machine use.

A machine-readable Prediction Output may include:

Prediction ID; Claim ID; Output Type; Output Value; Unit; Horizon; Boundary; Condition References; Methodology Reference; Formation Time; Output Version.

Human-Readable ≠ Machine-Readable

An executive dashboard sentence may be understandable to a person but insufficient for system interoperability.

A machine-readable record may be precise but unsuitable for human communication.

POFM may support both representations while preserving one governed meaning.

Output Interoperability

Where Prediction Outputs move across systems, the meaning of:

value; field; unit; horizon; boundary; state

must remain preserved.

Interoperable Format $\neq$ Interoperable Meaning by Default

Output Source

POFM should preserve the source of the final Prediction Output.

Possible sources include:

direct methodology output; transformed model output; selected candidate; human-adjusted output; aggregated output; hybrid output.

Output Provenance

Output Provenance identifies how the final Prediction Output emerged from earlier predictive components.

This supports later reconstruction.

Derived Prediction Output

An output may be derived from another predictive output.

Example:

Prediction A:

Demand = 12,000 units

Derived Prediction B:

Inventory shortage expected

These must remain distinguishable predictive outputs if they represent distinct claims.

Duplicate Prediction Output

The same prediction may appear in multiple systems.

Duplicate records should remain associated with the same Prediction Identity where appropriate.

Multiple Displays $\neq$ Multiple Independent Predictions

Output Conflict

An Output Conflict exists where multiple candidate final outputs cannot simultaneously represent the same Identified Predictive Claim without further governance.

Example:

Predicted demand: 12,000

and:

Predicted demand: 18,000

for the same object, horizon, boundary, and conditions.

The conflict must be:

resolved; separated; qualified; or preserved.

Claim–Output Conflict

A more fundamental conflict exists when the output does not represent the Identified Predictive Claim.

Example:

Claim:

Probability that system temperature exceeds 90°C

Output label:

System Failure Probability

PCIM may have correctly identified the claim.

POFM must prevent output formation from changing it.

Output Formation Gap

A Prediction Output Formation Gap exists where the Identified Predictive Claim cannot be transformed into a sufficiently governed final output.

Possible causes include:

unclear output form; missing unit; missing horizon; incompatible value and claim; unresolved candidate outputs; unjustified transformation; false precision; missing conditionality; semantic distortion.

Output Formation Restriction

POFM may restrict output formation where a material gap remains.

The predictive process may return to:

PCOM; PCIM; methodology; Prediction Basis; assumption governance
for clarification.

Unable to Form Output

POFM may conclude:

Unable to Form Governed Prediction Output

where the Predictive Claim cannot be represented without material distortion or unresolved ambiguity.

This is a legitimate governance outcome.

Output State

Possible Output States may include:

Candidate Output

Formed Output

Conditionally Formed Output

Restricted Output

Conflicting Output

Incomplete Output

Unable to Form

These states are specific to the output and should not be confused with the complete Prediction Formation State established under PSEM.

Output Version

Material changes to the formed Prediction Output should create a distinguishable version or historical state.

Examples:

Output V1: 12,000 units

Output V2: 10,500 units

The reason and lifecycle treatment of the later prediction-level change are governed elsewhere where applicable.

Historical Output Integrity

The output originally formed must remain reconstructable.

Later:

corrections; revised data; changed methodology; new assumptions; changed probability; revised predictions

must not silently rewrite the historical output.

Current Output $\neq$ Historical Formed Output

Output Correction vs Prediction Revision

A technical formatting error may be corrected without necessarily creating a prediction revision.

Example:

Output accidentally displayed:

120,000 units

instead of the correctly formed:

12,000 units

because of a UI formatting error.

This differs from changing the prediction itself.

Output Correction $\neq$ Prediction Revision by Default

The distinction must remain governable.

Output Formation and Probability

POFM may carry probability-related output fields where they are part of the Predictive Claim structure.

However, PPUS governs the legitimacy and representation of:

probability; uncertainty; confidence; ranges.

Prediction Output Formation $\neq$ Probability Governance

Output Formation and Feasibility

A Prediction Output may be fully formed even where PFES later determines that the predicted outcome is infeasible.

Formed Output $\neq$ Feasible Outcome

Output Formation and Communication

POFM establishes the governed Prediction Output.

PCRS later determines how it may be:

communicated; summarized; visualized; interpreted; relied upon.

A communication layer must not rewrite the governed output meaning.

Output Formation and Monitoring

PMRS may later act upon a prediction whose output changes because of revision or supersession.

POFM establishes the output associated with the relevant formation state.

Prediction Output Governance Sufficiency

Prediction Output Formation is sufficiently governed when:

the output represents the Identified Predictive Claim; output form is explicit; value/state/category is identifiable; unit is preserved where applicable; baseline is preserved where applicable; Horizon remains associated; Boundary remains associated; conditionality remains preserved; material transformations are traceable; human adjustments are identifiable where material; false precision is avoided; rounding does not create semantic distortion; human- and machine-readable forms preserve the same meaning; conflicts and formation gaps remain explicit; historical output states remain reconstructable.

Governance Requirements

POFM requires that:

1. every Governed Prediction Output remain associated with an Identified Predictive Claim; 2. output form be identifiable; 3. output value, category, range, threshold, direction, event, or timing state be explicit where applicable; 4. units remain identifiable where applicable; 5. applicable baseline remain identifiable; 6. Prediction Horizon remain associated with the output; 7. Predictive Boundary remain associated where material; 8. conditionality remain preserved; 9. scenario dependence remain preserved where applicable; 10. raw model output remain distinguishable from Governed Prediction Output; 11. intermediate outputs remain distinguishable from final output; 12. output selection remain traceable where materially relevant; 13. transformations remain traceable; 14. adjustments remain identifiable where material; 15. aggregation remain methodologically compatible; 16. normalization preserve meaning; 17. output precision remain methodologically supported; 18. rounding not silently change predictive meaning; 19. semantic integrity between claim and output be preserved; 20. human-readable and machine-readable representations preserve equivalent predictive meaning; 21. output source and provenance remain identifiable where material; 22. derived outputs remain distinguishable from source predictions; 23. duplicates not be treated as independent predictions; 24. Output Conflicts remain explicit; 25. Claim–Output Conflicts prevent silent formation; 26. Output Formation Gaps remain explicit; 27. inability to form an output remain an acceptable governance result; 28. material output changes remain historically distinguishable; 29. historical formed outputs remain reconstructable; 30. output correction remain distinguishable from prediction revision where relevant.

Governance Logic

POFM applies the following governance logic:

Governed Prediction Composition

Identified Predictive Claim

↓

Required Prediction Output Form Identified

↓

Output Value / State / Category / Range / Threshold Defined

↓

Units / Baseline Associated where applicable

↓

Prediction Horizon / Boundary Associated

↓

Conditions / Scenario Dependence Preserved

↓

Raw / Intermediate / Candidate Outputs Distinguished

↓

Selection / Transformation / Adjustment / Aggregation Applied where governed

↓

Semantic Integrity Against Predictive Claim Validated

↓

Precision / Rounding Assessed

↓

Human-Readable / Machine-Readable Representation Established where applicable

↓

Output Provenance Preserved

↓

Output Conflicts / Formation Gaps Assessed

↓

Prediction Output State Established

↓

Governed Prediction Output

↓

PSEM — Prediction State Establishment Module

Where sufficient output formation cannot be established:

Prediction Output Formation Gap → Clarification / Reformation / Qualification / Restricted Output / Unable to Form

Minimum Implementation Framework

A minimum implementation of POFM must establish:

A. Prediction Reference

The applicable prediction formation process.

B. Predictive Claim Reference

The Identified Predictive Claim established by PCIM.

C. Output Identifier

A unique or sufficiently distinguishable Prediction Output reference.

D. Output Type

At minimum, as applicable:

Point / Range / Threshold / Category / Direction / Event / Timing / Conditional / Structured / Other

E. Output Value / State

The actual prediction output.

F. Unit

Where applicable.

G. Comparison Baseline

Where applicable.

H. Prediction Horizon

The applicable future time point or interval.

I. Predictive Boundary

Where material.

J. Conditionality / Scenario Reference

Any conditions materially required to interpret the output.

K. Output Source

At minimum:

Direct / Selected / Transformed / Adjusted / Aggregated / Derived / Hybrid

L. Transformation / Adjustment Reference

Any material change applied before final output formation.

M. Precision / Rounding State

Where applicable.

N. Semantic Integrity State

At minimum:

Aligned / Conditionally Aligned / Conflicting / Unable to Determine

O. Output State

At minimum:

Candidate / Formed / Conditionally Formed / Restricted / Conflicting / Incomplete / Unable to Form

P. Output Version / Historical State

The applicable output version or equivalent historical record.

Governance Outputs

POFM may produce:

Governed Prediction Output; Prediction Output Record; Output Identifier; Output Type; output-value record; output-category record; output-range record; output-threshold record; output-direction record; output-event record; output-timing record; unit record; baseline record; Horizon association; Boundary association; conditionality record; scenario association; Output Source Record; transformation record; adjustment record; aggregation record; normalization record; precision record; rounding record; semantic-integrity assessment; human-readable output; machine-readable output; Output Provenance Record; Output Conflict Record; Claim–Output Conflict Record; Prediction Output Formation Gap; Output State; Output Version; historical output record.

The primary governed output is:

Governed Prediction Output

which becomes a principal input to:

PSEM — Prediction State Establishment Module.

Critical Distinctions

Predictive Claim $\neq$ Prediction Output Representation

Prediction Output $\neq$ Raw Model Output

Raw Model Output $\neq$ Governed Prediction Output

Conditional Claim $\neq$ Unconditional Output

Intermediate Output $\neq$ Final Prediction Output

Transformed Output $\neq$ Original Output

Output Normalization $\neq$ Meaning Change

Numeric Value Without Unit $\neq$ Complete Numeric Prediction Output

Same Output Value + Different Horizon $\neq$ Same Prediction Output

Output Representation $\neq$ Permission to Change Claim Meaning

Rounding $\neq$ Semantically Neutral by Default

Human-Readable $\neq$ Machine-Readable

Interoperable Format $\neq$ Interoperable Meaning by Default

Multiple Displays $\neq$ Multiple Independent Predictions

Current Output $\neq$ Historical Formed Output

Output Correction $\neq$ Prediction Revision by Default

Prediction Output Formation $\neq$ Probability Governance

Formed Output $\neq$ Feasible Outcome

Enterprise Application

POFM gives organizations a governed transition from:

what the prediction means

to:

what the prediction actually outputs.

This becomes critical where predictive systems use:

model scores; percentages; categories; dashboards; ranges; analyst adjustments; machine-readable interfaces; automated downstream systems.

Without Output Formation Governance, a correct Predictive Claim can still become distorted when represented.

POFM prevents:

a risk score from becoming a probability without justification; a conditional claim from becoming unconditional; a model output from being relabeled; a range from becoming a point estimate; units or horizons from disappearing; rounding from crossing a material threshold; an adjusted output from being represented as raw model output.

Use Case 1 — A Model Score Is Published as a Failure Probability

An industrial AI system produces:

Model Score: 0.84

The score measures similarity between current machine behavior and patterns historically associated with abnormal vibration.

A dashboard converts the result into:

84% probability of machine failure within 30 days.

The underlying model never produced that probability.

PCIM correctly identifies the actual claim as relating to:

future threshold exceedance in abnormal vibration indicators.

The critical governance question is:

Can a predictive system legitimately publish a failure probability merely because an internal model score happens to be represented on a 0–1 scale?

POFM requires the final output to preserve the actual Predictive Claim and prevents the score from being silently transformed into a different predictive meaning.

If a failure probability is later derived, that derivation must be separately governed.

Same Numerical Scale $\neq$ Same Predictive Meaning

Use Case 2 — Rounding Changes Whether a Prediction Crosses a Decision Threshold

A financial prediction produces:

49.6%

for a governed future condition.

The organization's operational rule triggers escalation when a displayed result reaches:

50%.

The user interface rounds 49.6% to:

50%.

The prediction is therefore escalated even though the underlying formed output remained below the operational threshold.

The critical governance question is:

Can a formatting decision legitimately change the operational meaning of a prediction?

POFM requires rounding and output representation to preserve material semantic and threshold relationships.

The organization may display:

50%

for readability only if the underlying governed value and threshold interpretation remain explicit and no downstream process falsely treats the rounded display as the original predictive output.

Rounding $\neq$ Semantically Neutral by Default

Canonical Boundary

POFM governs the form, value, category, range, threshold, direction, timing, unit, baseline, conditionality, normalization, transformation, selection, adjustment, aggregation, precision, semantic integrity, human- and machine-readable representation, provenance, state, versioning, and historical preservation of the Prediction Output representing an Identified Predictive Claim. It does not govern Prediction Composition itself, Predictive Claim Identification itself, overall Prediction Formation State, probability and uncertainty legitimacy, Prediction Feasibility, Collective Prediction composition, communication and reliance, prediction-level monitoring and revision, outcome determination, or predictive performance assessment.

POFM establishes:

what output legitimately represents the identified Predictive Claim.

It does not determine:

whether the complete prediction is ready to proceed as a fully formed Governed Prediction.

Canonical Closing Statement

A predictive claim can be correctly defined and still be corrupted by the way its output is formed.

POFM preserves the claim when it becomes a value, range, category, event, threshold, or machine-readable prediction output.

The final prediction output must represent the claim — not silently become a different prediction.

[↑](#)

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